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High-frequency line structure, subassembly, line card, and method for manufacturing line structure

Abstract

A high-frequency line structure includes: a high-frequency line substrate; ground lead pins fixed to ground ends provided in a bottom surface of the high-frequency line substrate; and signal lead pins fixed to signal line ends provided in the bottom surface of the high-frequency line substrate, wherein the signal lead pins are arranged between the ground lead pins, the signal lead pins have a structure in which each of the signal lead pins springs up in a direction toward a side on which the high-frequency line substrate is arranged, from a horizontal plane to which bottom surfaces of the ground lead pins pertains, and spring-up heights in the structure in which the respective signal lead pins spring up are substantially the same.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national phase entry of PCT Application No. PCT/JP2019/044530, filed on Nov. 13, 2019, which application is hereby incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a high-frequency line structure, a subassembly, and a line card

with excellent high-frequency characteristics, and a method for manufacturing a high-frequency line structure.

BACKGROUND

(3) A plurality of various optoelectronic components are densely mounted on a line card installed in communication devices to achieve desired communication functions. These communication devices are being developed for a wider band for transmission at 1 Tbps, and a wide band that enables transmission at 70 GHz or higher is becoming necessary in some regions of printed boards for each of a plurality of high-frequency lines on the printed boards that serve as substrates for the line card.

(4) However, with the miniaturization of optoelectronic components in recent years, the pitch of the plurality of high-frequency lines that propagate high-frequency signals between optoelectronic components is also becoming smaller and smaller, and this is causing crosstalk problems between adjacent high-frequency lines to become apparent. Crosstalk in such areas where the high-frequency lines and optoelectronic components are connected on the printed board has been an important issue.

(5) PTL 1 discloses a technology for miniaturizing a three-dimensional structure of an optical module in which an optical waveguide, an optoelectronic element, and so on, are mounted on a substrate. FIG. **12** shows a side cross-sectional view of an optical module **1001** described in PTL 1. An optical fiber **1002** is connected to the optical module **1001**, and a photodiode (PD) **1003** and a transimpedance amplifier (TIA) **1004** are mounted within a package. A high-frequency signal output from the TIA**1004** propagates to a high-frequency line **1005**, which penetrates a base that is made of an insulator, and a lead pin **1006** having the same thickness.

CITATION LIST

Patent Literature

(6) [PTL 1] Japanese Patent No 6122380.

SUMMARY

Technical Problem

(7) However, this optical module does not have a configuration for adjusting impedance matching, and crosstalk cannot be reduced. Consequently, it has been difficult to achieve a wider band that enables high-frequency signals to be stably propagated from DC to 70 GHz.

Means for Solving the Problem

(8) To solve the foregoing problem, a high-frequency line structure according to the embodiments of present invention includes: a high-frequency line substrate; ground lead pins fixed to ground ends provided in a bottom surface of the high-frequency line substrate; and signal lead pins fixed to signal line ends provided in the bottom surface of the high-frequency line substrate, wherein the signal lead pins are arranged between the ground lead pins, the signal lead pins have a structure in which each of the signal lead pins springs up in a direction toward a side on which the high-frequency line substrate is arranged, from a horizontal plane to which bottom surfaces of the ground lead pins pertains, and spring-up heights in the structure in which the respective signal lead pins spring up are substantially the same.

(9) A method for manufacturing a high-frequency line structure according to embodiments of the present invention is a method for manufacturing a high-frequency line structure by mounting, on a base jig, a thick lead frames including a plurality of lines, which are arranged at an interval, base end portions of the lines being integrally connected by a joint portion and formed in a substantial comb shape, a thin lead frame including a plurality of lines, which are arranged at an interval, base end portions of the lines being integrally connected by a joint portion and formed in a substantial comb shape, the thin lead frame having ridge portions in the lines, and a high-frequency line substrate having a transmission line in a bottom surface thereof, the method including steps of: arranging the lines of the thin lead frame between the lines of the thick lead frame above the base jig, and arranging the ridge portions of the thin lead frame so as to come into contact with a top

surface of the base jig; bringing ground ends provided in the bottom surface of the high-frequency line substrate and leading end portions of the lines of the thick lead frame into contact with each other; bringing signal line ends provided in the bottom surface of the high-frequency line substrate and leading end portions of the lines of the thin lead frame into contact with each other; pressing a top surface of a positioning pressure jig placed on a top surface of a joint portion of the thick lead frame and a top surface of a joint portion of the thin lead frame, and a top surface of the high-frequency line substrate; fixing, using a conductive material, and electrically connecting portions at which the ground ends provided in the bottom surface of the high-frequency line substrate and the leading end portions of the lines of the thick lead frame are brought into contact with each other, and portions at which the signal line ends provided in the bottom surface of the high-frequency line substrate and the leading end portions of the lines of the thin lead frames are brought into contact with each other; and cutting the thick lead frame and the thin lead frame.

Effects of Embodiments of the Invention

(10) According to embodiments of the present invention, it is possible to provide a method for manufacturing a high-frequency line structure that enables highly accurate positioning when high-frequency lines are electrically connected to mount constituent components, to provide a high-frequency line structure, a subassembly, and a line card with high-frequency characteristics that enable a reduction in crosstalk over a wide band, and to provide an electronic component and an optical module component that have wide-band characteristics for next-generation 1 Tbps and beyond.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A is a bird's eye perspective view that shows a step of arranging constituent components in a method for manufacturing a high-frequency line structure according to a first embodiment of the present invention.

(2) FIG. **1**B is a side perspective view that shows the step of arranging the constituent components in the method for manufacturing a high-frequency line structure according to the first embodiment of the present invention.

(3) FIG. **2**A is a bird's eye perspective view that shows a step of pressing the constituent components in the method for manufacturing a high-frequency line structure according to the first embodiment of the present invention.

(4) FIG. **2**B is a side perspective view that shows the step of pressing the constituent components in the method for manufacturing a high-frequency line structure according to the first embodiment of the present invention.

(5) FIG. **3** is a bird's eye perspective view that shows a step of cutting thick lead frames and thin lead frames in the method for manufacturing a high-frequency line structure according to the first embodiment of the present invention.

(6) FIG. **4**A is a bird's eye perspective view of a high-frequency line structure according to the first embodiment of the present invention.

(7) FIG. **4**B is a side perspective view of a high-frequency line structure according to the first embodiment of the present invention.

(8) FIG. **5** is a bird's eye perspective view of a subassembly according to the first embodiment of the present invention.

(9) FIG. **6**A is a top perspective view of the subassembly according to the first embodiment of the present invention.

(10) FIG. **6**A is a side perspective view of the subassembly according to the first embodiment of the present invention.

(11) FIG. **7** shows high-frequency characteristics of the subassembly that are obtained by calculation according to the first embodiment of the present invention.
(12) FIG. **8** is a bird's eye perspective view of a subassembly according to the second embodiment of the present invention.
(13) FIG. **9**A is a top perspective view of the subassembly according to the second embodiment of the present invention.
(14) FIG. **9**B is a side perspective view of the subassembly according to the second embodiment of the present invention.
(15) FIG. **10** shows calculation results of high-frequency characteristics of the subassembly that are obtained by calculation according to the second embodiment of the present invention.
(16) FIG. **11** is a conceptual diagram of a line card according to a third embodiment of the present invention.
(17) FIG. **12** is a conceptual diagram of an optical module of a conventional technology.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

First Embodiment

(18) The first embodiment of the present invention will be described with reference to the drawings. A method for manufacturing a high-frequency line structure **10**, and the high-frequency line structure **10** according to the first embodiment of the present invention will be described with reference to FIGS. **1**A to **4**.
(19) <Method for Manufacturing High-frequency Line Structure>
(20) FIG. **1**A is a bird's eye perspective view that shows an arrangement of constituent components in a method for manufacturing a high-frequency line structure according to the first embodiment of the present invention, and FIG. **1**B is a side perspective view. A high-frequency line substrate **111** according to the present embodiment is constituted by two metal layers, and the bottom and top surfaces each have coplanar lines (size: about 10 mm×10 mm). Thick lead frames **121**, which come into contact with ground ends of the coplanar lines provided in the bottom surface of the high-frequency line substrate **111**, and thin lead frames **131**, which come into contact with signal line ends of the coplanar lines, are prepared. A heat sink **112** is provided in the bottom surface of the high-frequency line substrate **111**. These constituent components are arranged on a base jig **22**, and the constituent components are aligned by positioning pressure jigs **21**.
(21) Each thick lead frame **121** (thickness: 0.3 mm) includes a plurality of lines **121**A (length: 20 mm, width: 0.2 mm), the plurality of lines **121**A are arranged at an interval (0.8 mm), and base end portions of the lines **121**A are integrally connected by a joint portion **121**B and formed in a substantial comb shape.
(22) Each thin lead frame **131** (thickness: 0.2 mm) includes a plurality of lines **131**A (length: 20 mm, width: 0.2 mm), the plurality of lines **131**A are arranged at an interval (0.8 mm), and base end portions of the lines **131**A are integrally connected by a joint portion **131**B and formed in a substantial comb shape. In FIG. **1**A, each thick lead frame **121** consists of four lines, and each thin lead frame **131** consists of three lines. However, the number of lines is not limited thereto. As will be described later, the substantially comb-shaped lines **131**A of the thin lead frames **131** need only be arranged between the substantially comb-shaped lines **121**A of the thin lead frames **121**.
(23) As shown in FIG. **1**B, each thin lead frame **131** has a ridge portion **15**A at a position 2.4 mm away from a leading end portion of each line **131**A on a surface in contact with a top surface **22**C of the base jig **22**. The ridge portion **15**A extends in a direction perpendicular to the lengthwise direction of each thin lead frame **131**. The ridge portion **15**A may have a bent shape or a curved shape. The ridge portion **15**A need only have a protruding shape so as to come into contact with the top surface **22**C of the base jig **22**. As a result of thus having the ridge portions **15**A, each thin lead frame **131** has a structure in which it springs up, and the height of a base end portion thereof (the height in the vertical direction from a horizontal plane with which the ridge portions **15**A are in contact to the top surface of the base end; hereinafter referred to as a “spring-up amount **16**”) is

about 10 mm.

(24) As shown in FIG. **1**B, each thin lead frame **131** has a ridge portion **15**B on a surface that comes into contact with a transmission line provided in the bottom surface of the high-frequency line substrate **111**. The ridge portion **15**B extends in a direction perpendicular to the lengthwise direction of the thin lead frame **131**. The ridge portion **15**B is formed such that a top surface of the leading end portion of each line **131**A of the thin lead frame **131** is in the same plane as a top surface of a leading end portion of each line **121**A of the thick lead frame **121**. Each ridge portion **15**B may have a bent shape or a curved shape.

(25) In the present embodiment, Kobar (iron-nickel-cobalt alloy) is used as a material of the thick lead frames **121** and the thin lead frames **131**. Here, the thermal expansion coefficient of Kobar is substantially equal to the thermal expansion coefficient of ceramic used in printed boards. Any other alloy of tungsten, iron, Ni, or the like can alternatively be used as the material of the thick lead frames **121** and the thin lead frames **131**.

(26) The substantially comb-shaped lines **131**A of each thin lead frame **131** are arranged between the substantially comb-shaped lines **121**A of the corresponding thick lead frame **121**. At this time, the lines **131**A of the thin lead frame **131** and the lines **121**A of the thick lead frame **121** are arranged so as to not interfere with each other.

(27) The leading end portions of the lines **131**A of the thin lead frame **131** and the leading end portions of the lines **121**A of the thick lead frame **121** are aligned in the lengthwise direction. In other words, the leading ends of the top surfaces of the lines **131**A of the thin lead frame **131** and the leading ends of the top surfaces of the lines **121**A of the thick lead frame **121** are arranged on a straight line with which a plane intersects that is perpendicular, in the lengthwise direction, to a horizontal plane to which the top surface of the thick lead frame **121** and the top surface of the leading end portion of the thin lead frame **131** pertain.

(28) The ridge portion **15**A of each thin lead frame **131** is arranged so as to come into contact with the top surface **22**C of the base jig **22**.

(29) The ground end of the transmission line provided in the bottom surface of the high-frequency line substrate **111** is brought into contact with the leading end portions of the lines **121**A of the thick lead frames **121**. The signal line end of the transmission line provided in the bottom surface of the high-frequency line substrate **111** is brought into contact with the leading end portions of the lines **131**A of each thin lead frame **131**.

(30) Each positioning pressure jig **21** has, on its bottom surface, a recessed portion **21**A to be fitted to the joint portion **131**B of the corresponding thin lead frame **131**, and protruding portions **21**B to be fitted to spaces between the lines **121**A of the corresponding thick lead frame **121**. The depth of the recessed portion **21**A is 0.15 mm, and it is desirable that the depth of the recessed portion **21**A is smaller than the thickness of each thin lead frame **131**. It is desirable that the width of the recessed portion **21**A is about 0.1 mm larger than the width of the joint portion **131**B of each thin lead frame **131**. The height of each protruding portion **21**B is 0.15 mm, and it is desirable that the height of the protruding portion **21**B is smaller than the thickness of each thick lead frame **121**. It is desirable that the width of each protruding portion **21**B is about 0.1 mm smaller than the width of each space between the lines **121**A of each thick lead frame **121**.

(31) The base jig **22** has, on the top surface **22**C, protruding portions **22**B to be fitted to spaces between the lines **121**A of the thick lead frames **121**. The height of each protruding portion **22**B is 0.15 mm, and it is desirable that the height of the protruding portion **22**B is smaller than the thickness of the thick lead frame **121**. It is desirable that the width of the protruding portion **22**B is about 0.1 mm smaller than the width of the space between the lines **121**A of the thick lead frame **121**.

(32) The base jig **22** also have protruding portions **22**A on the top surface **22**C, and the height of each protruding portion **22**A is 0.25 mm. It is desirable that the height of the protruding portions **22**A is smaller than the thickness of the thick lead frame **121**. The depth of the groove between the

protruding portions **22**A need only be a depth at which the heat sink **112** fits.

(33) FIGS. **2**A and **2**B show a state where the constituent components are pressed. After all the constituent components have been arranged and aligned on the base jig **22**, the top surfaces of the positioning pressure jigs **21** and the top surface of the high-frequency line substrate **111** are pressed. The pressing direction is indicated by arrows **23** in FIGS. **2**A and **2**B.

(34) Due to deforming due to springness in the lead lengthwise direction thereof, the thin lead frames **131** are fitted by the base jig **22** and the positioning pressure jigs **21**, and thus, firm fitting is achieved, rattling is suppressed, and the constituent components are integrated. In detail, when the positioning pressure jigs **21** and the high-frequency line substrate **111** are pressed, the thin lead frames **131** have elasticity, and a reaction force is generated in the upward direction against the high-frequency line substrate **111** with the ridge portions **15**A in contact with the top surface **22**C of the base jig **22** as a fulcrum. Thus, the adhesion between the thick lead frames **121**, the thin lead frames **131**, and the high-frequency line substrate **111** is improved.

(35) Further, the recessed portions **21**A of the bottom surfaces of the positioning pressure jigs **21** fit to the joint portions **131**B of the thin lead frames **131**, the protruding portions **21**B fit to the spaces between the lines **121**A of the thick lead frames **121**, and the protruding portions **22**B of the base jig **22** fit to the spaces between the lines **121**A of the thick lead frames **121**. As a result, it is possible to minimize gaps at portions at which the positioning pressure jigs **21**, the thin lead frames **131**, and the thick lead frames **121** are fitted to each other, achieve firm fitting, and position the constituent components while suppressing rattling.

(36) In this state, the constituent components are accurately fixed with high adhesion and electrically connected by a conductive bonding material (not shown), such as silver solder, gold tin solder, conductive resin, or a conductive adhesive, at portions at which the top surfaces of the leading end portions of the thin lead frames **131** and of the thick lead frames **121** are in contact with a metal surface provided on the bottom surface of the high-frequency line substrate **111**.

(37) After being firmly fixed, the lead frames are simultaneously cut at positions about 1 mm to 2 mm away from the high-frequency line substrate **111** by cutting blades **32**, as shown in FIG. **3**. At this time, the thick lead frames **121**, which are arranged below the thin lead frames **131**, prevent the lines **131**A of the thin lead frames **131** from deforming downward during the cutting.

(38) In the present embodiment, the lead frames are cut at positions about 1 mm to 2 mm away from the high-frequency line substrate **111**, but the cutting positions are not limited thereto. However, if the distance from the high-frequency line substrate **111** is 2 mm or longer, the fixation using solder during the aforementioned mounting will become difficult.

(39) <Configuration of High-Frequency Line Structure>

(40) FIG. **4**A is a bird's eye perspective view of the high-frequency line structure **10** in a finished state after removed from the base jig **22**, and FIG. **4**B is a side perspective view. Ground lead pins **122**, which are formed from the thick lead frames **121**, signal lead pins **132**, which are formed from the thin lead frames **131**, the high-frequency line substrate **111**, and the heat sink **112** are integrated, and the high-frequency line structure **10** has a three-dimensional structure.

(41) The high-frequency line substrate **111** has a metal two-layer structure, has coplanar lines in the bottom and top surfaces, and has a dielectric material **142** therebetween. In the coplanar lines in the bottom surface, opening portions are provided at portions of a bottom ground **141**, which is made of metal, and bottom signal lines are provided in these opening portions. In the coplanar lines in the top surface, opening portions are provided at portions of a top ground **143**, which is made of metal, and top signal lines are provided in these opening portions. Bottom signal line ends **144** are provided at high-frequency line substrate ends on the bottom side, are connected to the top surface side of the high-frequency line substrate **111** by pseudo coaxial lines **145**, and are then connected to top signal line ends **146**.

(42) The ground lead pins **122** and the signal lead pins **132** are cut at the same positions during the above-described manufacturing process, and therefore have an equal length, which is about 3 mm.

Further, the signal lead pins **132** and the ground lead pins **122** have a structure characterized in that a thickness **171** of each ground lead pin **122** is larger than a thickness **172** of each signal lead pin **132**.
(43) Each signal lead pins **132** has a ridge portion **132**A, which corresponds to the ridge portion **15**A of the thin lead frame **131** in the above-described manufacturing process. Further, each signal lead pins **132** have a structure in which the signal lead pin **132** springs up from a horizontal plane to which the bottom surface of the ground lead pins **122** pertains. A spring-up height **16**H in the spring-up structure is the distance between a horizontal plane to which the bottom surface of each ground lead pin **122** pertains and the top surface of a leading end of each signal lead pin **132**. The spring-up height **16**H is about 0.25 mm to 0.3 mm, and it is desirable that the spring-up height **16**H is smaller than or equal to the height of each ground lead pin **122**.
(44) Each signal lead pin **132** has a ridge portion **132**B at a leading end portion on the side connected to the high-frequency line substrate **111**, at a position about 2 mm from the leading end on a surface that comes into contact with the transmission line provided in the bottom surface of the high-frequency line substrate **111**. The ridge portion **132**B extends in a direction perpendicular to the lengthwise direction of the signal lead pin **132**. The ridge portion **132**B is formed such that the top surface of the leading end portion of the signal lead pin **132** is in the same plane as the top surface of the leading end portion of each ground lead pin **122**. As a result, when the top surfaces of the signal lead pins **132** are connected to the bottom surfaces of the signal line ends of the coplanar lines by a conductive bonding material such as solder, the connection becomes easier and stronger. Each ridge portion **132**B may have a bent shape or a curved shape.
(45) As shown in FIG. **4**B, each signal lead pins **132** is characterized in the structure in which the signal lead pin **132** springs up from the horizontal plane to which the bottom surface of the corresponding ground lead pin **122** pertains, and has the spring-up height **16**H. The spring-up height **16**H is the distance between the horizontal plane to which the bottom surface of the ground lead pin **122** pertains and the top surface of the leading end of the signal lead pin **132**. As shown in FIGS. **1**B and **2**B, the spring-up height **16**H depends on the spring-up amount **16** employed in the process of manufacturing the high-frequency line structure **10**, and depends on the length (line length) in the lengthwise direction of the thin lead frame **131**, and can therefore be sufficiently controlled by adjusting the line length at the time of designing.
(46) Here, in the process of manufacturing the high-frequency line structure **10**, the spring-up amount **16** can be unified since the base ends of the lines **131**A of each thin lead frame **131** is integrated at the joint portion **131**B. As a result, substantially the same spring-up height **16**H can be applied to all the signal lead pins **132**, and thus, when the signal lead pins **132** are mounted on a printed board, variation in high-frequency characteristics between the signal lead pins can be sufficiently suppressed.
(47) Furthermore, during the aforementioned pressing using the positioning pressure jigs **21**, the reaction force against the high-frequency line substrate **111** depends on the spring-up amount **16**. Therefore, the adhesiveness between the high-frequency line substrate **111**, the thick lead frames **121**, and the thin lead frames **131** can be controlled by controlling the spring-up amount **16** through the line length.
(48) <Configuration of Subassembly>
(49) FIG. **5** shows a substrate **5***o* (hereinafter referred to as a “subassembly”) on which the above-described high-frequency line structure **10** is mounted. A printed board **41** has a bottom ground **42** on a bottom surface, and coplanar lines on a top surface. These coplanar lines have top grounds that include top ground ends **43**, and signal lines that include signal line ends **44**.
(50) The bottom surfaces of the ground lead pins **122** and the top surfaces of the top ground ends **43**, and the bottom surfaces of the signal lead pins **132** and the top surfaces of the signal line ends **44** are connected at connecting portions **51** and connecting portions **52**, respectively, by an electrically conductive bonding material such as solder. Here, since each signal lead pin **132** has the

spring-up height **16**H as mentioned above, a predetermined space is provided between the bottom surface of the signal lead pin **132** and the top surface of the corresponding signal line end **44**. Thus, mechanical strength can be ensured by the conductive bonding material such as solder, and stable high-frequency characteristics can be provided.

(51) On the top surface of the high-frequency line substrate **111**, a wide-band amplifier element **61** is flip-chip mounted between electrodes **62**, the ground ends and the signal line ends of the coplanar lines on the top surface of the high-frequency line substrate **111**. Note that denotation of a DC terminal, and so on, of the wide-band amplifier element is omitted in the drawings.

(52) FIG. **6**A is a top perspective view of the subassembly **50**, and FIG. **6**B is a side perspective view. The signal lead pins **132** are arranged between the ground lead pins **122**. The ground lead pins **122** and the signal lead pins **132** are electrically connected to the ground ends and the signal line ends, respectively, of the coplanar lines in the bottom surface of the high-frequency line substrate **111**. In FIG. **6**B, regions **6** (hereinafter referred to as “shielding regions”) in which a shielding effect is achieved by the ground lead pins **122** are indicated by dotted lines.

(53) The subassembly **50** according to the present embodiment has a structure characterized in that the thickness **171** of each ground lead pin **122** is larger than the thickness **172** of each signal lead pin **132**. When the thickness of the ground lead pin is smaller than or equal to the thickness of the signal lead pin **132**, the signal lead pin in principle has a structure in which an electromagnetic field distribution can easily spread, since air is present in the area around the inclined structure portion of the signal lead pin, the impedance in this area becomes high impedance, degrading reflection loss, and at the same time, crosstalk problems due to the large electromagnetic field distribution spread is likely to appear.

(54) In the present embodiment, since the thickness **171** of the ground lead pin **122** is larger than the thickness **172** of the signal lead pin **132**, the ground lead pin **122** not only realizes suppression of high impedance due to an increase in capacitance between the ground (earth) and the signal lead pin **132**, but also suppresses crosstalk between the signal lead pins **132** by functioning as shield plates for electromagnetic fields. For example, computational simulations confirmed that this configuration is especially effective when the thickness of the ground lead pin **122** is 1.5 times or more the thickness of the signal lead pin **132**.

(55) The upper limit of the thickness of the ground lead pin **122** is determined by the balance between the effect of improving crosstalk suppression and the effect of reducing reflection loss. As mentioned above, as the thickness of the ground lead pin **122** increases, the function thereof as a shield plate is further improved and the effect of crosstalk suppression increases. Meanwhile, the increase in capacitance promotes lower impedance, and impedance mismatch occurs in the shielding regions **6** in FIG. **6**B. Accordingly, it is desirable that the ground lead pin **122** has a thickness that can suppress crosstalk and can also suppress impedance mismatch.

(56) If the thickness of the thick lead frame **121**, which corresponds to the thickness of the ground lead pin **122**, becomes thicker than 0.5 mm, burrs may occur at the cutting portions after the cutting in the above-described manufacturing process, and subsequent electrical connection by means of solder or the like may become defective. For this reason, the upper limit of the thickness of the ground lead pin **122** is 0.5 mm.

(57) FIG. **7** shows the results of simulating high-frequency characteristics when the thicknesses of the signal lead pin **132** and the ground lead pin **122** are 0.2 mm and 0.3 mm, respectively. An output signal and inter-channel crosstalk were calculated by inputting a high-frequency signal to the subassembly **50** in which a high-frequency IC was mounted on the high-frequency line structure **10** of this embodiment that has the signal lead pins **132** and ground lead pins **122** described above. Commercially available software “ANSYS HFSS” (Ansys) was used for the calculation.

(58) The characteristics obtained by the calculation are those between the signal line ends of the coplanar lines provided on the printed board, and include the characteristics of the high-frequency

IC mounted on the high-frequency line structure **10**. The characteristics that rise to the right indicate the inter-channel crosstalk between adjacent channels. A dotted line graph **71** indicates inter-channel crosstalk when the signal lead pin and the ground lead pin have the same structure (a structure having a spring-up structure with a thickness of 0.2 mm), and a solid line graph **72** indicates inter-channel crosstalk in the present embodiment. A solid line graph **73** indicates band characteristics. It is indicated that the quality of high frequency signals is maintained up to 70 GHz. An improvement in crosstalk of about 20 dB was calculated over the entire frequency band, and the effectiveness of the present embodiment is confirmed. Accordingly, a high-frequency line structure **10** that has low crosstalk characteristic in a wide band is provided.

Second Embodiment

(59) Next, the second embodiment of the present invention will be described. FIG. **8** shows a subassembly **60** according to the second embodiment. The subassembly **60** is substantially the same as the subassembly **50** of the first embodiment, but is different in that high-frequency lines formed on a top surface of a high-frequency line substrate **113** include coplanar lines **147** that allow opposing connections at both ends of the substrate, and differential microstrip lines **45** are provided on the top surface of the printed board. This high-frequency line structure **10** realizes a three-dimensional high-frequency line structure without short-circuiting orthogonal high-frequency lines.

(60) <Configuration of Subassembly>

(61) FIG. **9**A is a top perspective view of the subassembly **60**, and FIG. **9**B is a side perspective view. The signal lead pins **132** are arranged between the ground lead pins **122**, and the signal lead pins **132** and the ground lead pins **122** have a structure characterized in that the thickness **171** of each ground lead pin **122** is larger than the thickness **172** of each signal lead pin **132** in the shielding regions **6** that are in contact with the high-frequency line substrate **113**. Since air is present around the inclined structure of the signal lead pins, the signal lead pins in principle have a structure in which the electromagnetic field distribution can easily spread, causing the impedance in that area to become high impedance and degrading the reflection loss, while at the same time, crosstalk problems due to the large electromagnetic field distribution spread is likely to appear.

(62) In the present embodiment, the ground lead pins **122** can not only suppress high impedance by due to an increase in the capacitance between the ground (earth) and the signal lead pins **132**, but also suppress crosstalk between the signal lead pins **132** by functioning as shield plates for electromagnetic fields. For example, computational simulations confirmed that this configuration is especially effective when the thickness of each ground lead pin **122** is 1.5 times or more the thickness of each signal lead pin **132**.

(63) Meanwhile, the upper limit of the thickness of the ground lead pin **122** is determined by the balance between the effect of improving suppression of crosstalk suppression and the effect of reducing reflection loss. As mentioned above, as the thickness of the ground lead pin **122** increases, the function thereof as a shield plate is further improved and the effect of crosstalk suppression increases. Meanwhile, the increase in capacitance promotes lower impedance, and impedance mismatch occurs in the shielding regions **6** in FIG. **6**B. Accordingly, it is desirable that the ground lead pin **122** has a thickness that can suppress crosstalk and also suppress impedance mismatch.

(64) If the thickness of the thick lead frame **121**, which corresponds to the thickness of the ground lead pin **122**, becomes thicker than 0.5 mm, burrs may occur at the cutting portions after the cutting in the above-described manufacturing process, and subsequent electrical connection by means of solder or the like may become defective. For this reason, the upper limit of the thickness of the ground lead pin **122** is 0.5 mm.

(65) FIG. **10** shows the results of simulating high-frequency characteristics when the thicknesses of the signal lead pin **132** and the ground lead pin **122** are 0.2 mm and 0.3 mm, respectively. An output signal and inter-channel crosstalk were calculated by inputting a high-frequency signal to the subassembly **60** in which a high-frequency IC was mounted on the high-frequency line structure **10** of this embodiment that has the signal lead pins **132** and ground lead pins **122**

described above. Commercially available software "ANSYS HFSS" (Ansys) was used for the calculation.
(66) The characteristics obtained by the calculation are those between the signal line ends of the coplanar lines provided on the printed board. The characteristics that rise to the right indicate the inter-channel crosstalk between adjacent channels. A dotted line trace **81** indicates inter-channel crosstalk when the signal lead pin and the ground lead pin have the same structure (a structure having a spring-up structure with a thickness of 0.2 mm), and a solid line graph **82** indicates inter-channel crosstalk in the present embodiment. A solid line **83** indicates band characteristics. It is indicated that the quality of high frequency signals is maintained up to 70 GHz. An improvement in crosstalk of about 20 dB was calculated over the entire frequency band, and the effectiveness of the present embodiment is confirmed. Accordingly, a high-frequency line structure **10** that has low crosstalk characteristic in a wide band is provided by providing the structure of the present embodiment.
Third Embodiment
(67) Next, the third embodiment of the present invention will be described.
(68) <Configuration of Line Card>
(69) FIG. **11** shows an example of a line card, which is the third embodiment of the present invention. Input ports **93**, PDs **94**, TIAs **95**, and output ports **96** are mounted together with the subassemblies **92** of the first embodiment on a substrate **91**. Optical fibers (not shown) are connected to the input ports, and high frequency optical signal are input thereto. The input optical signals are converted to high-frequency electrical signals via the PDs **94** and the TIAs **95**, subjected to signal processing by the subassemblies **92**, and are output from the output ports **96**. The line card of the present embodiment enables high-frequency signal processing with low crosstalk characteristics in a wide band.
(70) Although the subassembly of the first embodiment is used as the subassembly in the present embodiment, the subassembly of the second embodiment may alternatively be used, and any subassembly having the high-frequency line structure of embodiments of the present invention has the same effect.
(71) The line card according to the present embodiment is for reception, but may alternatively be for transmission, and a laser diode (LD) and a light emitting diode (LED) may be mounted as optoelectronic devices.
(72) In the first to third embodiments of the present invention, coplanar lines are used as the transmission lines in the high-frequency line substrates in and **113** and the printed board **41**, but any other transmission lines such as microstrip lines may alternatively be used.
(73) Although the dimensions of the constituent portions, components, and the like in the high-frequency line structure, the subassembly, the line card, and the method for manufacturing a high-frequency line structure according to the first to third embodiments of the present invention have been described, the dimensions are not limited thereto, and need only be dimensions with which the constituent portions, components, and the like function.
(74) Although the dielectric material that constitutes the high-frequency line substrate is low-loss ceramic such as alumina in all the embodiments, it is needless to say that aluminum nitride, zirconia, cozilite, zircon, forsterite, quartz glass, or the like can be used instead. Further, in all the embodiments, gold plating for the purpose of improving wettability of solder is applied in general to each line connecting portion when the lead pins are electrically connected by means of solder. However, gold plating is not the essence of the present invention, and is therefore not specifically described.
INDUSTRIAL APPLICABILITY
(75) Embodiments of the present invention can be applied to a high-frequency line structure, a subassembly, and a line card with excellent high-frequency characteristics, as well as an electronic component and an optical module component having wide-band characteristics for next-generation

1 Tbps and beyond.

REFERENCE SIGNS LIST

(76) **10** High-frequency line structure **111**, **113** High-frequency line substrate **121** Thick lead frame **131** Thin lead frame **122** Ground lead pin **132** Signal lead pin **144** Bottom coplanar line end **146** Top coplanar line end **147** Coplanar line **15**A, **15**B Ridge portion **16** Flip-up amount **21** Positioning pressure jig **22** Base jig **32** CUTTING BLADE.

Claims

1. A high-frequency line structure comprising: a high-frequency line substrate; ground lead pins, end portions of the ground lead pins being fixed to respective ground ends in a bottom surface of the high-frequency line substrate; and signal lead pins, first end portions of the signal lead pins being fixed to respective signal line ends in the bottom surface of the high-frequency line substrate, wherein each of the signal lead pins includes, in addition to the first end portion, a second end portion, and a ridge portion between the first end portion and the second end portion, and wherein each of the signal lead pins has a structure in which the signal lead pin extends downward from the first end portion to the ridge portion, bends at the ridge portion, and extends upward from the ridge portion to the second end portion.
2. The high-frequency line structure according to claim 1, wherein each of the ground lead pins is thicker than each of the signal lead pins.
3. The high-frequency line structure according to claim 1, wherein a thickness of each of the ground lead pins is 1.5 times or more a thickness of each of the signal lead pins.
4. The high-frequency line structure according to claim 1, wherein a thickness of each of the ground lead pins is 0.5 mm or less.
5. The high-frequency line structure according to claim 1, wherein spring-up heights in the structure from which respective ones of the signal lead pins spring up are the same.
6. The high-frequency line structure according to claim 1, wherein the ridge portion is configured to protrude toward a direction away from the high-frequency line substrate, and extend in a direction perpendicular to a lengthwise direction of each of the signal lead pins.
7. A subassembly comprising: the high-frequency line structure according to claim 1, wherein the signal lead pins are arranged between the ground lead pins; and a printed board, wherein a bottom surfaces of the ground lead pins are fixed to top surfaces of top ground ends of the printed board, and wherein a bottom surface of the signal lead pins are fixed to signal line ends of the printed board.
8. A line card comprising a subassembly, the subassembly comprising: a high-frequency line structure comprising: a high-frequency line substrate; ground lead pins, end portions of the ground lead pins being fixed to respective ground ends in a bottom surface of the high-frequency line substrate; and signal lead pins, first end portions of the signal lead pins being fixed to respective signal line ends in the bottom surface of the high-frequency line substrate, wherein each of the signal lead pins includes, in addition to the first end portion, a second end portion, and a ridge portion between the first end portion and the second end portion, and wherein each of the signal lead pins has a structure in which the signal lead pin extends downward from the first end portion to the ridge portion, bends at the ridge portion, and extends upward from the ridge portion to the second end portion; and a printed board, wherein a bottom surface of the ground lead pins are fixed to top surfaces of top ground ends of the printed board, and wherein a bottom surface of the signal lead pins are fixed to signal line ends of the printed board.
9. A method for manufacturing a high-frequency line structure, the method comprising: providing a thick lead frame including a first plurality of lines, the first plurality of lines being arranged at an interval, base end portions of the first plurality of lines being integrally connected by a first joint portion and arranged in a comb shape; providing a thin lead frame including a second plurality of

lines, the second plurality of lines being arranged at an interval, base end portions of the second plurality of lines being integrally connected by a second joint portion and arranged in a comb shape, each of the second plurality of lines having first end portions, ridge portions, and the base end portions in the lines, the ridge portions being between the first end portions and the base end portions, each of the second plurality of lines bends at the ridge portion; providing a high-frequency line substrate having a transmission line in a bottom surface thereof; mounting, on a base jig, the thick lead frame, the thin lead frame, and the high-frequency line substrate; arranging the second plurality of lines of the thin lead frame between the first plurality of lines of the thick lead frame above the base jig; arranging the ridge portions of the thin lead frame so as to come into contact with a top surface of the base jig; bringing ground ends in the bottom surface of the high-frequency line substrate and leading end portions of the first plurality of lines of the thick lead frame into contact with each other; bringing signal line ends in the bottom surface of the high-frequency line substrate and leading end portions of the second plurality of lines of the thin lead frame into contact with each other, each of the second plurality of lines of the thin lead frame extend downward from the leading end portion to the ridge portion, bends at the ridge portion, and extends upward from the ridge portion the base end portion; pressing a top surface of a positioning pressure jig placed on a top surface of the first joint portion of the thick lead frame and a top surface of the second joint portion of the thin lead frame, and a top surface of the high-frequency line substrate; fixing, using a conductive material, and electrically connecting portions at which the ground ends provided in the bottom surface of the high-frequency line substrate and the leading end portions of the first plurality of lines of the thick lead frame are brought into contact with each other, and portions at which the signal line ends provided in the bottom surface of the high-frequency line substrate and the leading end portions of the second plurality of lines of the thin lead frame are brought into contact with each other; and cutting the thick lead frame and the thin lead frame.

10. The method for manufacturing the high-frequency line structure according to claim 9, wherein a recessed portion to be fitted to the second joint portion of the thin lead frame and a protruding portion to be fitted to a space between the lines of the thick lead frame are provided in one surface of the positioning pressure jig.